American International University-Bangladesh

Course Title: Software Engineering

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Project Title

School Management System

1. Introduction:

1.1 Brief Overview of the Project

The **School Management System (SMS)** is a software application designed to automate and simplify the daily administrative and academic activities of a school. The system provides a centralized platform where administrators, teachers, students, and parents can access relevant information according to their roles. It reduces paperwork, minimizes human errors, and improves communication among all stakeholders.

1.2 About the System

The proposed system will manage student records, teacher information, class schedules, attendance, examination results, fee payments, and notices. Different users will have separate login accounts with role-based access to ensure security and privacy.

1.3 Why are We Developing the System?

Many schools still rely on manual record keeping using paper files or spreadsheets. This approach is time-consuming, prone to errors, and makes retrieving information difficult. The School Management System aims to digitize these processes, making school operations more efficient, organized, and reliable.

2. Problem Statement:

2.1 Existing Problems

The current manual management system faces several challenges, including:

- Manual maintenance of student and teacher records.
- Difficulty in tracking attendance.
- Time-consuming result preparation.
- Errors in fee management.
- Slow retrieval of information.
- Risk of data loss due to paper-based records.

- Lack of centralized data management.

2.2 Motivation

The motivation behind this project is to develop an affordable and user-friendly system that simplifies school administration and improves productivity.

2.3 Problem Being Solved

The proposed system solves the following problems:

- Automates school management tasks.
- Reduces paperwork.
- Improves data accuracy.
- Enhances communication.
- Provides secure data storage.
- Generates reports quickly.

2.4 Why is the Project Necessary?

The project is necessary because educational institutions require efficient management systems to handle increasing amounts of academic and administrative data while ensuring accuracy, security, and fast access.

3. Project Objectives:

Main Objectives

- Develop a centralized school management system.
- Reduce manual paperwork.
- Improve administrative efficiency.
- Ensure secure data management.
- Generate reports automatically.

Specific Objectives

- Student registration management.
- Teacher management.
- Class and subject management.

- Attendance management.
- Examination and result management.
- Fee management.
- Notice management.
- Role-based authentication.
- Report generation.
- Password management.

4. Methodology:

Requirement Gathering Method

Requirements will be collected through:

- Interviews with teachers and school administrators.
- Observation of current school activities.
- Reviewing existing school management procedures.
- Literature review of similar systems.

Development Approach

The project will follow the **Waterfall Software Development Model**, where each phase is completed before moving to the next.

Development Phases

1. Requirement Analysis
2. System Design
3. Database Design
4. Implementation
5. Testing
6. Deployment
7. Maintenance

Programming Language

- C#

Framework

- .NET Framework (Windows Forms)

IDE

- Microsoft Visual Studio 2022

Database

- Microsoft SQL Server 2022

Technologies Used

- C#
- Windows Forms
- SQL Server
- ADO.NET
- SQL Server Management Studio (SSMS)

5. System Requirements:

Features and Functionality

Admin

- Login
- Manage students
- Manage teachers
- Manage classes
- Manage subjects
- Manage fees
- Publish notices
- View reports

Teacher

- Login

- View assigned classes
- Mark attendance
- Upload marks
- View student information

Student

- Login
- View profile
- View attendance
- View examination results
- View notices

Functional Requirements

- User authentication.
- Student registration.
- Teacher registration.
- Attendance recording.
- Examination management.
- Result generation.
- Fee collection.
- Notice management.
- Report generation.
- Password change.

Non-Functional Requirements

- User-friendly interface.
- Fast response time.
- Data security.

- High reliability.
- Easy maintenance.
- Data backup.
- Scalability.
- System availability.

6. Technologies to be Used:

Software Development Process Model

Waterfall Model

Why Waterfall Model?

The Waterfall model is suitable because:

- Requirements are clearly defined.
- Easy to understand.
- Simple documentation.
- Suitable for academic projects.
- Easy progress tracking.
- Each phase has clear deliverables.

Model Description

The project progresses through sequential phases:

Requirement Analysis → System Design → Implementation → Testing → Deployment → Maintenance

7. Project Timeline:

Week	Activities
Week 1	Initial planning and topic selection
Week 2	Requirement gathering and feasibility study

Week	Activities
Week 3	Software Requirement Specification (SRS) preparation
Week 4	System design (Use Case Diagram, ER Diagram, DFD, UML)
Week 5	Database design and table creation
Week 6	User Interface (UI) design
Week 7	Admin module development
Week 8	Student module development
Week 9	Teacher module development
Week 10	Attendance and examination modules
Week 11	Fee management and report generation
Week 12	Integration of all modules
Week 13	Testing and bug fixing
Week 14	Final documentation and presentation

8. Budget Estimation (Optional):

Item	Estimated Cost (BDT)
Internet	1,000
Electricity	500
Printing & Binding	1,000
Software Tools	Free (Visual Studio Community & SQL Server Express)
Miscellaneous	500
Total Estimated Cost	**3,000 BDT**

9. Conclusion:

The **School Management System** is designed to automate the administrative and academic activities of a school by replacing manual processes with a secure, efficient, and user-friendly digital solution. The system will improve record management, reduce paperwork, minimize errors, and provide quick access to information for administrators, teachers, and students. Using **C# Windows Forms** and **Microsoft SQL Server**, the project will deliver a reliable application while demonstrating the complete software engineering process, including requirement analysis, system design, implementation, testing, and documentation. This project is practical, achievable within a semester, and provides valuable experience in designing and developing a real-world information system.